



North Sydney Girls High School

2025

HSC TRIAL EXAMINATION

Mathematics Extension 2

**General
Instructions**

- Reading Time – 10 minutes
- Working Time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- For questions in Section II, show relevant mathematical reasoning and/or calculations

**Total marks:
100**

Section I – 10 marks (pages 2 – 7)

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 8 – 15)

- Attempt Questions 11 – 16
- Allow about 2 hours and 45 minutes for this section

NAME: _____

TEACHER: _____

STUDENT NUMBER:

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Question	1–10	11	12	13	14	15	16	Total
Mark	/10	/14	/15	/15	/16	/14	/16	/100

Section I

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section

Use the multiple choice answer sheet for Questions 1–10.

- 1 Consider the statement:

“If you do not try, then you will not succeed.”

Which of the following statements is logically equivalent to the statement above?

- A. If you did not succeed, then you tried.
- B. If you did not succeed, then you did not try.
- C. If you succeeded, then you tried.
- D. If you succeeded, then you did not try.

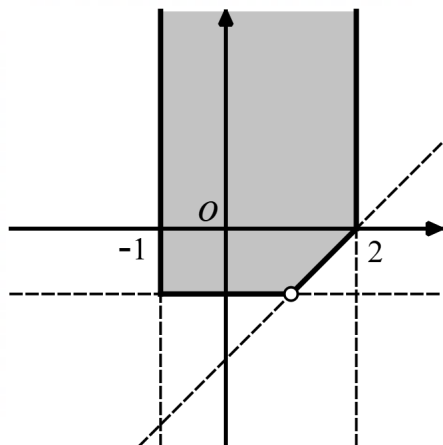
- 2 Which of the following vectors is perpendicular to the vector $2\vec{i} + 9\vec{j} - 3\vec{k}$?

- A. $3\vec{i} + \vec{j} - 5\vec{k}$
- B. $6\vec{i} - \vec{j} - \vec{k}$
- C. $-\vec{i} + 2\vec{j} - 3\vec{k}$
- D. $-3\vec{i} + 2\vec{j} + 4\vec{k}$

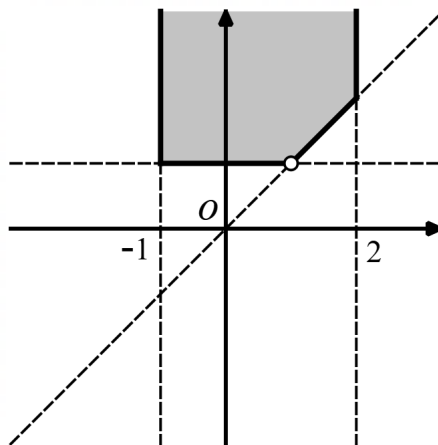
- 3 Let R be the region in the complex plane defined by $-2 \leq z + \bar{z} \leq 4$ and $\frac{\pi}{4} \leq \text{Arg}(z - 1 + i) \leq \pi$.

Which diagram best represents the region R ?

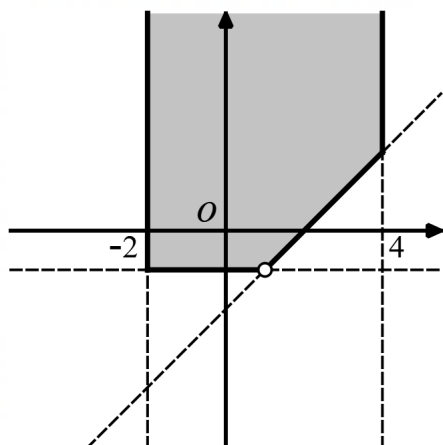
A.



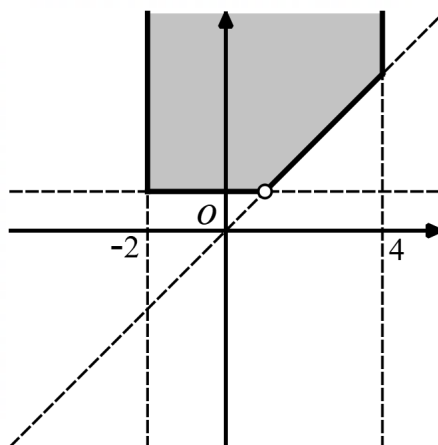
B.



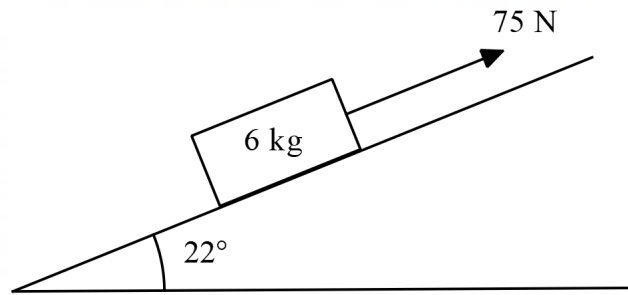
C.



D.



- 4 An object of mass 6 kg is placed on a smooth ramp that is inclined at an angle of 22° to the horizontal. It is pulled up the ramp by a force of 75 Newtons as shown in the diagram.



The acceleration due to gravity is 9.8 ms^{-2} .

Which of the following is closest to the acceleration of the block?

- A. 3.4 ms^{-2}
B. 8.8 ms^{-2}
C. 16.2 ms^{-2}
D. 21.6 ms^{-2}
- 5 Which of the following is a vector equation of the line joining the points $A(3, -2, 5)$ and $B(-1, 4, 9)$?

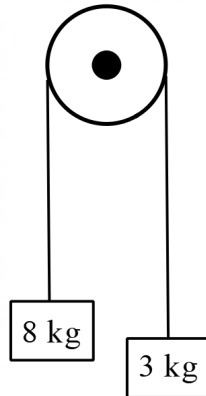
A. $\vec{r} = \begin{pmatrix} -1 \\ 4 \\ 9 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ -2 \\ 5 \end{pmatrix}$

B. $\vec{r} = \begin{pmatrix} -1 \\ 4 \\ 9 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -3 \\ -2 \end{pmatrix}$

C. $\vec{r} = \begin{pmatrix} 3 \\ -2 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 4 \\ 9 \end{pmatrix}$

D. $\vec{r} = \begin{pmatrix} 2 \\ -3 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ -2 \\ 5 \end{pmatrix}$

- 6 A light inextensible string passes over a smooth pulley. Attached to the ends of the string are masses of 8 kg and 3 kg as shown.



The acceleration due to gravity is 9.8 ms^{-2} .

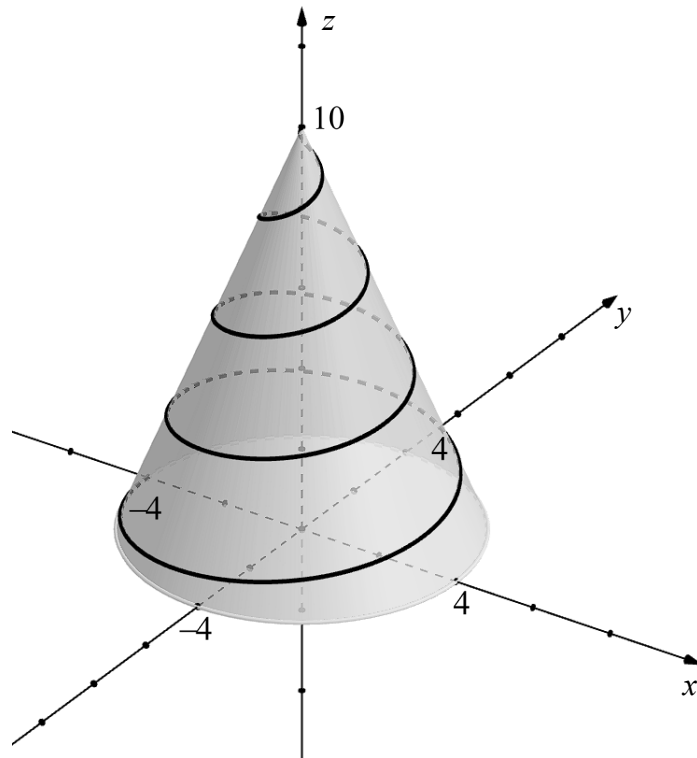
Which of the following is closest to the acceleration of the masses?

- A. 4.5 ms^{-2}
B. 6.1 ms^{-2}
C. 13.5 ms^{-2}
D. 21.6 ms^{-2}
- 7 Which of the following is equal to $\int_0^{\frac{\pi}{4}} \frac{1}{6 + \cos 2x} dx$?

- A. $\int_0^1 \frac{1}{7 + 5t^2} dt$
B. $\int_0^1 \frac{1}{5 + 7t^2} dt$
C. $\frac{1}{3} \int_0^1 \frac{1}{3t^2 + t + 3} dt$
D. $\frac{1}{3} \int_0^1 \frac{1}{3t^2 - t + 3} dt$

- 8 A particle undergoes simple harmonic motion between $x = -3$ and 5 where x is the displacement measured in metres from a fixed point O . Given that its maximum acceleration is 20 ms^{-2} , what is the velocity of the particle at $x = 0$?
- A. 8 ms^{-1}
B. $5\sqrt{3} \text{ ms}^{-1}$
C. $4\sqrt{5} \text{ ms}^{-1}$
D. $\frac{8\sqrt{15}}{3} \text{ ms}^{-1}$
- 9 Of the following expressions, which one need **NOT** contain a term involving arctan in its anti-derivative?
- A. $\frac{2}{x^2 + 2x + 7}$
B. $\frac{e^x}{e^{2x} + 5e^x + 6}$
C. $\frac{1}{\sqrt{x}(x+3)}$
D. $\frac{2 \sin x}{\cos 2x + 9}$

- 10 The diagram below shows a curve on the surface of a cone with radius 4 and height 10.



Which of the following position vectors can **NOT** describe the curve?

- A. $\underline{r}(t) = 2\left(t - \frac{19}{8}\right)\sin(-4\pi t)\underline{i} + 2\left(t - \frac{19}{8}\right)\cos(4\pi t)\underline{j} + 5\left(t - \frac{3}{8}\right)\underline{k}$ for $\frac{3}{8} \leq t \leq \frac{19}{8}$
- B. $\underline{r}(t) = 2(t-2)\cos(4\pi t)\underline{i} + 2(t-2)\sin(4\pi t)\underline{j} + 5t\underline{k}$ for $0 \leq t \leq 2$
- C. $\underline{r}(t) = 2(t-2)\cos(4\pi t)\underline{i} + 2(t-2)\sin(-4\pi t)\underline{j} + 5t\underline{k}$ for $0 \leq t \leq 2$
- D. $\underline{r}(t) = 2(t-2)\cos(4\pi t)\underline{i} + 2(2-t)\sin(-4\pi t)\underline{j} + 5t\underline{k}$ for $0 \leq t \leq 2$

Section II

90 marks

Attempt Questions 11–16

Allow about 2 hours and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (14 marks) Use a SEPARATE writing booklet

- (a) Consider the complex numbers $w = 1 - 2i$ and $z = 1 + 3i$.
- (i) Express $w + \bar{z}$ in the form $x + iy$, where x and y are real numbers. 2
- (ii) Express $\frac{w}{z}$ in modulus-argument form. 2
- (b) Find $\int xe^{-x} dx$. 2
- (c) Find the angle between the vectors $\underline{u} = 2\underline{i} - 5\underline{j} + \underline{k}$ and $\underline{v} = -\underline{i} + 4\underline{j} + 3\underline{k}$, giving the angle in degrees correct to 1 decimal place. 3
- (d) Given that $3 - \sqrt{3}i$ is a fourth root of $-72(1 + \sqrt{3}i)$, find the remaining 3 roots and express them in the form $re^{i\theta}$. 2
- (e) Consider the sequence u_n where $u_1 = 0$, $u_{n+1} = 2n - u_n$ for $n = 1, 2, \dots$. 3
Use mathematical induction to show that $2u_n = 2n - 1 + (-1)^n$ for $n = 1, 2, \dots$.

Question 12 (15 marks) Use a SEPARATE writing booklet

- (a) Consider the Statement A. 2

Statement A: ‘If the last digit of the integer n is 5 then n is divisible by 5.’

Prove by contradiction that Statement A is true.

- (b) Use an appropriate substitution of the form $x = a + b \sin \theta$ to evaluate 3

$$\int_7^{10} \sqrt{(x-4)(10-x)} \, dx.$$

- (c) The velocity of a particle moving in a straight line is given by $v^2 = 128 + 32x - 16x^2$, where v is measured in ms^{-1} and x is the displacement measured in m from the origin.

- (i) Show that the particle moves in simple harmonic motion with period $\frac{\pi}{2}$ s. 2

- (ii) What distance does the particle travel during a full period of its motion? 2

- (d) The position vectors of points A and B are $\begin{pmatrix} -7 \\ -3 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 5 \\ -1 \\ -8 \end{pmatrix}$ respectively. 3

Find the vector equation of the sphere with AB as its diameter.

- (e) (i) Prove that $\forall x, y \in \mathbb{R}^+, \frac{x+y}{2} \geq \sqrt{xy}$. 1

- (ii) Let a, b, c and d be positive real numbers. Use the result from (i) to prove that 2

$$(a+b+c+d) \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d} \right) \geq 16.$$

Question 13 (15 marks) Use a SEPARATE writing booklet

(a) Evaluate $\int_0^{\pi} x \sin x \cos^4 x \, dx$ using the property $\int_0^a f(x) \, dx = \int_0^a f(a-x) \, dx$. 2

(b) Consider the two lines in three dimensions given by 3

$$\vec{r} = \begin{pmatrix} -2 \\ 5 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 3 \\ -6 \end{pmatrix} \text{ and } \vec{r} = \begin{pmatrix} 5 \\ 4 \\ -1 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ -5 \\ 8 \end{pmatrix}.$$

Find the point of intersection of the two lines.

(c) The acceleration of a particle moving in a straight line is given by $a = 4 - v^2$, where v is its velocity.

Initially, the particle has a velocity of $\frac{2}{3} \text{ ms}^{-1}$ at $x = 0$.

(i) What is the displacement of the particle when it is travelling at a velocity of 1 ms^{-1} ? 2

(ii) What is the velocity of the particle at $t = \frac{1}{2}$? 4

(d) The integral I_n is defined by $I_n = \int_0^1 e^{-x} (1-x)^n \, dx$ for $n \geq 0$.

(i) Show that $I_n = 1 - nI_{n-1}$ for $n \geq 1$. 2

(ii) Given that I_n can be expressed in the form $a + be^{-1}$ where a and b are integers, find an expression for b in terms of n . 2

Question 14 (16 marks) Use a SEPARATE writing booklet

(a) Prove by mathematical induction that $n^3 - 1 > n^2 + n$, for integers $n \geq 2$. 3

(b) Use the method of partial fractions to find the value of a , where $a > 0$, such that 4

$$\int_0^a \frac{3-5x}{(1+x)(1-x+2x^2)} dx = 0.$$

(c) Let $w = \cos \theta + i \sin \theta$, where $0 < \theta < \pi$.

It is known that the complex number $w^2 + \frac{3\sqrt{2}}{w} - 3$ is purely imaginary.

(i) Show that $2 \cos^2 \theta + 3\sqrt{2} \cos \theta - 4 = 0$. 2

(ii) Hence, or otherwise, find w . 2

(iii) A and B are two points on an Argand diagram representing two distinct complex numbers z_1 and z_2 respectively. Suppose that $z_2 = wz_1$ and $|z_1| = 3$. 2
Find the exact length of AB .

(d) Consider $f(x) = x^m - mx$ where $m > 1$. 3
By using calculus, or otherwise, show that $x^m - 1 \geq m(x - 1)$ for all $x > 0$.

Question 15 (14 marks) Use a SEPARATE writing booklet

- (a) Let z and w be non-zero complex numbers. Consider the statements below:

3

$$P: z\bar{w} + \bar{z}w = 0$$

$$Q: \frac{z}{w} = ik, k \in \mathbb{R}$$

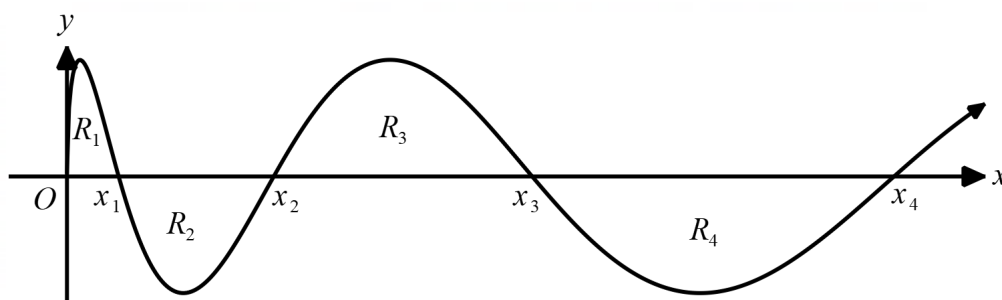
Prove that $P \Rightarrow Q$.

- (b) (i) By using a suitable substitution, show that

3

$$\int \sin \sqrt{x} \, dx = 2 \sin \sqrt{x} - 2\sqrt{x} \cos \sqrt{x} + C.$$

The diagram below shows part of the curve $y = \sin \sqrt{x}$ for $x \geq 0$.



The curve intersects the x -axis at $0, x_1, x_2, x_3, x_4, \dots$

The regions bounded by the curve and the x -axis are defined by $R_1, R_2, R_3, R_4, \dots$ as shown in the diagram.

- (ii) By first finding an expression for x_n , or otherwise, express the area of the region R_n in terms of n and π .

3

Question 15 continues on page 13

Question 15 (continued)

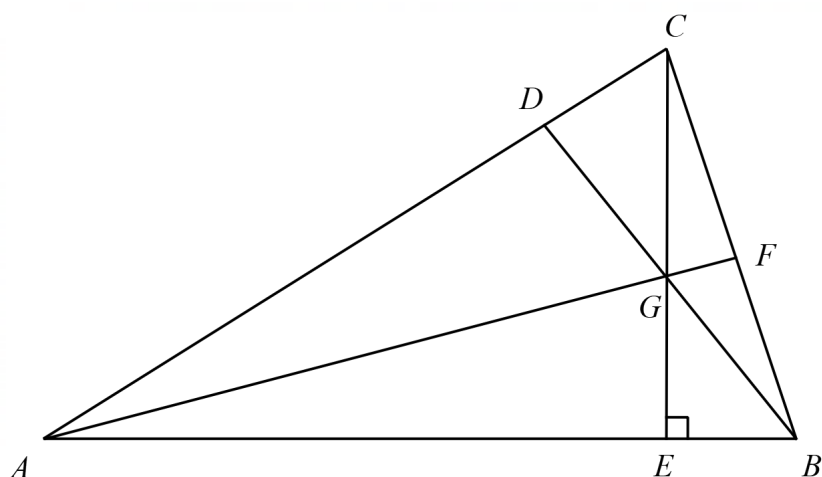
- (c) The diagram shows triangle ABC .

The point E lies on AB so that $AE : EB = \lambda : 1$ and $CE \perp AB$.

The point G is the mid-point of CE .

BG is produced to meet AC at D and $AD : DC = r : 1$.

The point F is chosen on BC so that A, G, F are collinear.



NOT TO
SCALE

Let $\overrightarrow{BE} = \underline{u}$ and $\overrightarrow{EG} = \underline{v}$.

- (i) Express $\overrightarrow{BG}$ in terms of $\underline{u}$ and $\underline{v}$.

1

- (ii) By considering $\overrightarrow{BD}$, use vector methods to show that $r = \lambda + 1$.

4

End of Question 15

Question 16 (16 marks) Use a SEPARATE writing booklet

- (a) It is given that for positive numbers $a_1, a_2, a_3, \dots, a_n$ with arithmetic mean A ,

$$A^n \geq a_1 \times a_2 \times a_3 \times \dots \times a_n. \quad (\text{Do NOT prove this.})$$

Let k be a positive integer.

- (i) Prove that $k^k \geq 1 \times 3 \times 5 \times \dots \times (2k-1)$. **2**

- (ii) Hence prove that $(k^2 + k)^k \geq (2k)!$. **2**

- (b) Consider the equation $(z+2)^8 = z^8$.

- (i) Show that the 6 non-real roots of the equation can be expressed in the form **4**

$$-1 - i \cot\left(\frac{k\pi}{8}\right), \quad k = \pm 1, \pm 2, \pm 3.$$

- (ii) Hence, find the value of $\sin^2\left(\frac{\pi}{8}\right)\sin^2\left(\frac{2\pi}{8}\right)\sin^2\left(\frac{3\pi}{8}\right)$ given that the product of roots of the equation is -16 . **3**

Question 16 continues on page 15

Question 16 (continued)

(c) Let $z = \cos \theta + i \sin \theta$.

(i) Find the four values of z such that $\operatorname{Im}(z^2 + \bar{z}) = 0$. **2**

Let z_1 and z_2 be two of the values of z obtained in (i) such that $\operatorname{Im}(z_1) < 0 < \operatorname{Im}(z_2)$.

Also let $S_n = \sum_{r=1}^n w^r$, where $w = \frac{z_2}{z_1}$, for any positive integer n .

(ii) Find all positive integers k such that $(S_n)^k + (S_{n+1})^k + (S_{n+2})^k = 2$ for any positive integer n . **3**

End of paper



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Question	1–10	11	12	13	14	15	16	Total
Mark	/10	/14	/15	/15	/16	/14	/16	/100

Section I

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section

Use the multiple choice answer sheet for Questions 1–10.

- 1 Consider the statement:

“If you do not try, then you will not succeed.”

Which of the following statements is logically equivalent to the statement above?

- A. If you did not succeed, then you tried.
- B. If you did not succeed, then you did not try.
- C. If you succeeded, then you tried.
- D. If you succeeded, then you did not try.

Consider

P : Do not try

Q : Do not succeed

The statement is $P \Rightarrow Q$.

The contrapositive is logically equivalent i.e. $\sim Q \Rightarrow \sim P$

- 2 Which of the following vectors is perpendicular to the vector $2\hat{i} + 9\hat{j} - 3\hat{k}$?

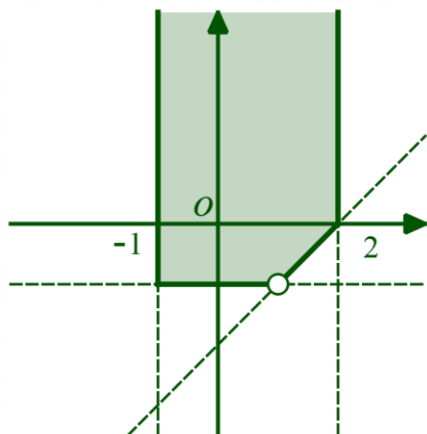
- A. $3\hat{i} + \hat{j} - 5\hat{k}$
- B. $6\hat{i} - \hat{j} - \hat{k}$
- C. $-\hat{i} + 2\hat{j} - 3\hat{k}$
- D. $-3\hat{i} + 2\hat{j} + 4\hat{k}$

$$(2\hat{i} + 9\hat{j} - 3\hat{k}) \cdot (-3\hat{i} + 2\hat{j} + 4\hat{k}) = 2(-3) + 9(2) - 3(4) = 0$$

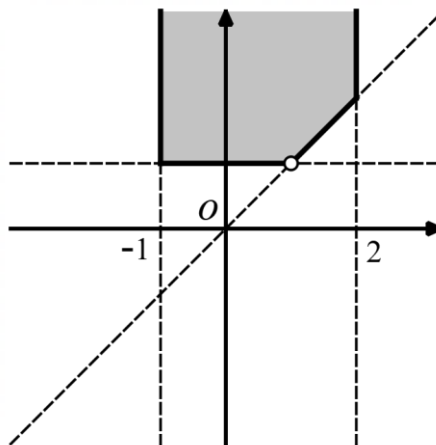
- 3 Let R be the region in the complex plane defined by $-2 \leq z + \bar{z} \leq 4$ and $\frac{\pi}{4} \leq \text{Arg}(z - 1 + i) \leq \pi$.

Which diagram best represents the region R ?

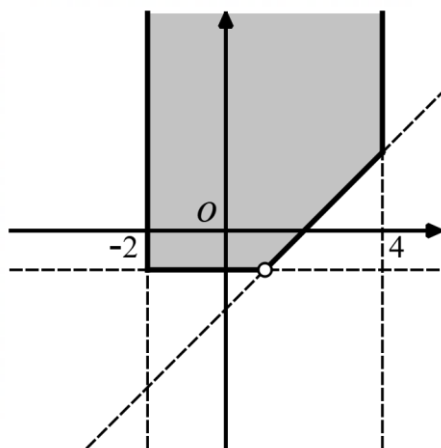
A.



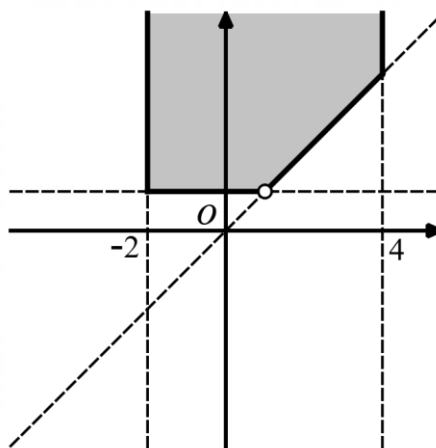
B.



C.



D.



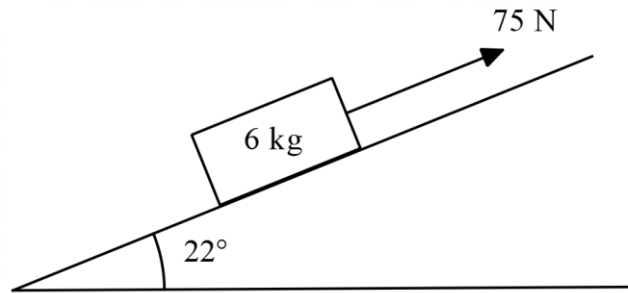
$$-2 \leq z + \bar{z} \leq 4$$

$$-2 \leq 2\text{Re}(z) \leq 4$$

$$-1 \leq \text{Re}(z) \leq 2$$

$$\frac{\pi}{4} \leq \text{Arg}(z - (1 - i)) \leq \pi$$

- 4 An object of mass 6 kg is placed on a smooth ramp that is inclined at an angle of 22° to the horizontal. It is pulled up the ramp by a force of 75 Newtons as shown in the diagram.



The acceleration due to gravity is 9.8 ms^{-2} .

Which of the following is closest to the acceleration of the block?

- A. 3.4 ms^{-2}
- B. 8.8 ms^{-2}
- C. 16.2 ms^{-2}
- D. 21.6 ms^{-2}

$$\begin{aligned}F_{\text{net}} &= 75 - 6 \times 9.8 \times \sin 22^\circ \\6a &= 75 - 6 \times 9.8 \times \sin 22^\circ \\a &= \frac{75 - 6 \times 9.8 \times \sin 22^\circ}{6} \\a &\approx 8.8\end{aligned}$$

- 5 Which of the following is a vector equation of the line joining the points $A(3, -2, 5)$ and $B(-1, 4, 9)$?

A. $\vec{r} = \begin{pmatrix} -1 \\ 4 \\ 9 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ -2 \\ 5 \end{pmatrix}$

B. $\vec{r} = \begin{pmatrix} -1 \\ 4 \\ 9 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -3 \\ -2 \end{pmatrix}$

C. $\vec{r} = \begin{pmatrix} 3 \\ -2 \\ 5 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 4 \\ 9 \end{pmatrix}$

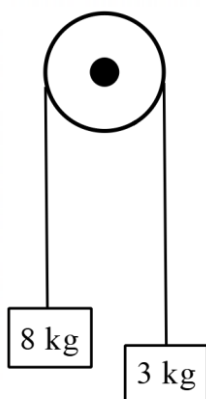
D. $\vec{r} = \begin{pmatrix} 2 \\ -3 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ -2 \\ 5 \end{pmatrix}$

Direction vector:

$$\begin{pmatrix} 3 - (-1) \\ -2 - 4 \\ 5 - 9 \end{pmatrix} = \begin{pmatrix} 4 \\ -6 \\ -4 \end{pmatrix}$$

Direction vector can be any vector parallel to $\begin{pmatrix} 4 \\ -6 \\ -4 \end{pmatrix}$ e.g. $\begin{pmatrix} 2 \\ -3 \\ -2 \end{pmatrix}$

- 6 A light inextensible string passes over a smooth pulley. Attached to the ends of the string are masses of 8 kg and 3 kg as shown.



The acceleration due to gravity is 9.8 ms^{-2} .

Which of the following is closest to the acceleration of the masses?

- A. 4.5 ms^{-2}
- B. 6.1 ms^{-2}
- C. 13.5 ms^{-2}
- D. 21.6 ms^{-2}

For the 8 kg mass:

$$8a = 8g - T \quad (1)$$

For the 3 kg mass:

$$3a = T - 3g \quad (2)$$

(1) + (2):

$$11a = 5g$$

$$a = \frac{5g}{11}$$

$$a \approx 4.5$$

7 Which of the following is equal to $\int_0^{\frac{\pi}{4}} \frac{1}{6 + \cos 2x} dx$?

A. $\int_0^1 \frac{1}{7 + 5t^2} dt$

B. $\int_0^1 \frac{1}{5 + 7t^2} dt$

C. $\frac{1}{3} \int_0^1 \frac{1}{3t^2 + t + 3} dt$

D. $\frac{1}{3} \int_0^1 \frac{1}{3t^2 - t + 3} dt$

Using the substitution $t = \tan x$

$$\frac{dt}{dx} = \sec^2 x$$

$$\frac{dt}{1+t^2} = dx$$

$$= \int_0^1 \frac{1}{6 + \cos 2x} dx$$

$$= \int_0^1 \frac{1}{6 + \frac{1-t^2}{1+t^2}} \frac{dt}{1+t^2}$$

$$= \int_0^1 \frac{1}{6(1+t^2) + 1 - t^2} dt$$

$$= \int_0^1 \frac{1}{7 + 5t^2} dt$$

- 8 A particle undergoes simple harmonic motion between $x = -3$ and 5 where x is the displacement measured in metres from a fixed point O . Given that its maximum acceleration is 20 ms^{-2} , what is the velocity of the particle at $x = 0$?

- A. 8 ms^{-1}
B. $5\sqrt{3} \text{ ms}^{-1}$
C. $4\sqrt{5} \text{ ms}^{-1}$
D. $\frac{8\sqrt{15}}{3} \text{ ms}^{-1}$

Centre: $\frac{-3+5}{2} = 1$

Amplitude: $\frac{5-(-3)}{2} = 4$

$$\therefore \ddot{x} = -n^2(x-1)$$

Maximum acceleration occurs at left most extremity i.e. $x = -3$

$$20 = -n^2(-3-1)$$

$$n^2 = 5$$

$$v^2 = n^2(A^2 - (x-1)^2)$$

At $x = 0$

$$v^2 = 5(4^2 - (0-1)^2)$$

$$v^2 = 75$$

$$v = \pm 5\sqrt{3}$$

- 9 Of the following expressions, which one need **NOT** contain a term involving arctan in its anti-derivative?

A. $\frac{2}{x^2 + 2x + 7}$

B. $\frac{e^x}{e^{2x} + 5e^x + 6}$

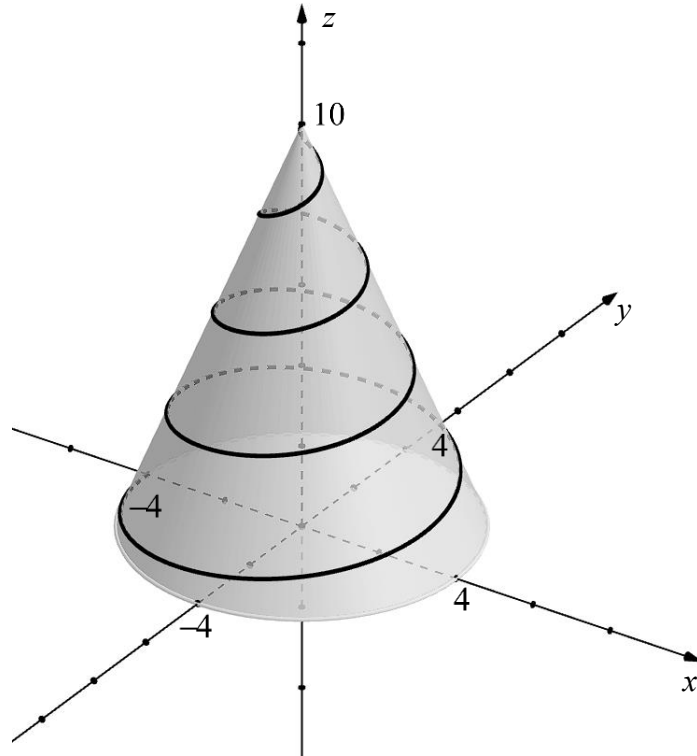
C. $\frac{1}{\sqrt{x}(x+3)}$

D. $\frac{2 \sin x}{\cos 2x + 9}$

$$\frac{e^x}{(e^x + 2)(e^x + 3)} = \frac{e^x}{e^{2x} + 5e^x + 6} = \frac{e^x}{e^{2x} + 5e^x + \left(\frac{5}{2}\right)^2 - \frac{1}{4}} = \frac{e^x}{\left(e^x + \frac{5}{2}\right)^2 - \frac{1}{4}}$$

As the denominator is not in the form $[f(x)]^2 + a^2$, it will not contain a term involving arctan.

- 10 The diagram below shows a curve on the surface of a cone with radius 4 and height 10.



Which of the following position vectors can **NOT** describe the curve?

- A. $\underline{r}(t) = 2\left(t - \frac{19}{8}\right)\sin(-4\pi t)\underline{i} + 2\left(t - \frac{19}{8}\right)\cos(4\pi t)\underline{j} + 5\left(t - \frac{3}{8}\right)\underline{k}$ for $\frac{3}{8} \leq t \leq \frac{19}{8}$
- B. $\underline{r}(t) = 2(t-2)\cos(4\pi t)\underline{i} + 2(t-2)\sin(4\pi t)\underline{j} + 5t\underline{k}$ for $0 \leq t \leq 2$
- C. $\underline{r}(t) = 2(t-2)\cos(4\pi t)\underline{i} + 2(t-2)\sin(-4\pi t)\underline{j} + 5t\underline{k}$ for $0 \leq t \leq 2$
- D. $\underline{r}(t) = 2(t-2)\cos(4\pi t)\underline{i} + 2(2-t)\sin(-4\pi t)\underline{j} + 5t\underline{k}$ for $0 \leq t \leq 2$

All options starts at $(-4, 0, 0)$ and ends at $(0, 0, 10)$, so these two points along cannot allow us to determine the answer.

Check for direction of “rotation” by checking location after $\frac{1}{4}$ of a cycle.

$$T = \frac{2\pi}{4\pi} = \frac{1}{2}$$

$$\frac{1}{4}T = \frac{1}{8}$$

For C, when $t = \frac{1}{8}$, $\underline{j}$ component gives:

$$2\left(\frac{1}{8} - 2\right)\sin\left(-4\pi\left(\frac{1}{8}\right)\right) = 3.75 \neq 0$$

$\therefore$ C is not correct.

Section II

90 marks

Attempt Questions 11–16

Allow about 2 hours and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (14 marks) Use a SEPARATE writing booklet

(a) Consider the complex numbers $w = 1 - 2i$ and $z = 1 + 3i$.

(i) Express $w + \bar{z}$ in the form $x + iy$, where x and y are real numbers. 2

$$1 - 2i + 1 - 3i = 2 - 5i$$

(ii) Express $\frac{w}{z}$ in modulus-argument form. 2

$$\frac{1 - 2i}{1 + 3i} = \frac{1 - 2i}{1 + 3i} \times \frac{1 - 3i}{1 - 3i} = \frac{-5 - 5i}{10} = \frac{1}{2}(-1 - i) = \frac{\sqrt{2}}{2} \left(\cos\left(-\frac{3\pi}{4}\right) + i \sin\left(-\frac{3\pi}{4}\right) \right)$$

(b) Find $\int x e^{-x} dx$. 2

$$\int x e^{-x} dx = - \int x d(e^{-x}) = - \left(x e^{-x} - \int e^{-x} dx \right) = -x e^{-x} - e^{-x} + C$$

(c) Find the angle between the vectors $\underline{u} = 2\underline{i} - 5\underline{j} + \underline{k}$ and $\underline{v} = -\underline{i} + 4\underline{j} + 3\underline{k}$, giving the angle in degrees correct to 1 decimal place. 3

$$|\underline{u}| = \sqrt{2^2 + (-5)^2 + 1^2} = \sqrt{30}$$

$$|\underline{v}| = \sqrt{(-1)^2 + 4^2 + 3^2} = \sqrt{26}$$

$$\underline{u} \cdot \underline{v} = (2)(-1) + (-5)(4) + (1)(3) = -19$$

$$\underline{u} \cdot \underline{v} = |\underline{u}| |\underline{v}| \cos \theta$$

$$-19 = \sqrt{30} \sqrt{26} \cos \theta$$

$$\cos \theta = \frac{-19}{\sqrt{780}}$$

$$\theta \approx 132.9^\circ$$

- (d) Given that $3 - \sqrt{3}i$ is a fourth root of $-72(1 + \sqrt{3}i)$, find the remaining 3 roots and express them in the form $re^{i\theta}$.

2

$$3 - \sqrt{3}i = 2\sqrt{3}e^{i\left(-\frac{\pi}{6}\right)}$$

Since the roots are evenly distributed around a circle, the angle between each root must be

$$\frac{2\pi}{4} = \frac{\pi}{2}$$

The remaining roots are

$$\begin{aligned} & 2\sqrt{3}e^{i\left(-\frac{\pi}{6} + \frac{\pi}{2}\right)}, 2\sqrt{3}e^{i\left(-\frac{\pi}{6} + \pi\right)}, 2\sqrt{3}e^{i\left(-\frac{\pi}{6} + \frac{3\pi}{2}\right)} \\ & = 2\sqrt{3}e^{i\left(\frac{2\pi}{3}\right)}, 2\sqrt{3}e^{i\left(\frac{\pi}{3}\right)}, 2\sqrt{3}e^{i\left(\frac{5\pi}{6}\right)} \end{aligned}$$

- (e) Consider the sequence u_n where $u_1 = 0$, $u_{n+1} = 2n - u_n$ for $n = 1, 2, \dots$.

3

Use mathematical induction to show that $2u_n = 2n - 1 + (-1)^n$ for $n = 1, 2, \dots$.

Prove that it is true for $n = 1$

When $n = 1$,

Recurrence relation gives: $u_1 = 0$

Formula gives:

$$2u_1 = 2(1) - 1 + (-1)^1$$

$$2u_1 = 0$$

$$u_1 = 0$$

$\therefore$ it is true for $n = 1$

Assume that it is true for $n = k$

$$2u_k = 2k - 1 + (-1)^k$$

Prove that it is true for $n = k + 1$

$$\text{i.e. RTP: } 2u_{k+1} = 2(k+1) - 1 + (-1)^{k+1}$$

From recurrence relation:

$$u_{k+1} = 2k - u_k \quad (\text{given recurrence relation})$$

$$2u_{k+1} = 4k - 2u_k$$

$$2u_{k+1} = 4k - (2k - 1 + (-1)^k) \quad (\text{by assumption})$$

$$2u_{k+1} = 4k - 2k + 1 - (-1)^k$$

$$2u_{k+1} = 2k + 1 + (-1)(-1)^k$$

$$2u_{k+1} = 2(k+1) - 1 + (-1)^{k+1}$$

$\therefore$ it is true for $n = k + 1$ if it is true for $n = k$

Hence the formula is true by mathematical induction.

Question 12 (15 marks) Use a SEPARATE writing booklet

- (a) Consider the Statement A.

2

Statement A: 'If the last digit of the integer n is 5 then n is divisible by 5.'

Prove by contradiction that Statement A is true.

Assume that the last digit of the integer n is 5 and n is not divisible by 5.

As the last digit of n is 5, n can be expressed as:

$$n = 10k + 5 \text{ where } k \text{ is an integer}$$

$$n = 5(2k + 1)$$

$$n = 5A \text{ where } A = 2k + 1 \text{ is also an integer since } k \text{ is an integer}$$

$\therefore n$ is divisible by 5 which contradicts the assumption

Hence Statement A is true by contradiction.

- (b) Use an appropriate substitution of the form $x = a + b \sin \theta$ to evaluate

3

$$\int_7^{10} \sqrt{(x-4)(10-x)} \, dx.$$

$$\int_7^{10} \sqrt{(x-4)(10-x)} \, dx = \int_7^{10} \sqrt{-((x-7)+3)((x-7)-3)} \, dx = \int_7^{10} \sqrt{3^2 - (x-7)^2} \, dx$$

Using the substitution $x = 7 + 3 \sin \theta$

$$\frac{dx}{d\theta} = 3 \cos \theta \quad \text{when } x = 10, \theta = \frac{\pi}{2}$$

$$dx = 3 \cos \theta \, d\theta \quad \text{when } x = 7, \theta = 0$$

$$\int_0^{\frac{\pi}{2}} \sqrt{[(7+3 \sin \theta)-4][10-(7+3 \sin \theta)]} 3 \cos \theta \, d\theta$$

$$= \int_0^{\frac{\pi}{2}} 3 \cos \theta \sqrt{(3+3 \sin \theta)(3-3 \sin \theta)} \, d\theta$$

$$= \int_0^{\frac{\pi}{2}} 3 \cos \theta \sqrt{9-9 \sin^2 \theta} \, d\theta$$

$$= \int_0^{\frac{\pi}{2}} 9 \cos^2 \theta \, d\theta$$

$$= \frac{9}{2} \int_0^{\frac{\pi}{2}} 1 + \cos 2\theta \, d\theta$$

$$= \frac{9}{2} \left[\theta + \frac{1}{2} \sin 2\theta \right]_0^{\frac{\pi}{2}}$$

$$= \frac{9}{2} \left(\frac{\pi}{2} + \frac{1}{2} \sin \left(2 \times \frac{\pi}{2} \right) - \left(0 + \frac{1}{2} \sin (2 \times 0) \right) \right)$$

$$= \frac{9\pi}{4}$$

- (c) The velocity of a particle moving in a straight line is given by $v^2 = 128 + 32x - 16x^2$, where v is measured in ms^{-1} and x is the displacement measured in m from the origin.

- (i) Show that the particle moves in simple harmonic motion with period $\frac{\pi}{2}$ s. 2

$$\frac{1}{2}v^2 = 64 + 16x - 8x^2$$

$$\frac{d}{dx}\left(\frac{1}{2}v^2\right) = 16 - 16x$$

$$a = -4^2(x-1)$$

$$\therefore n = 4$$

$$T = \frac{2\pi}{n} = \frac{2\pi}{4} = \frac{\pi}{2}$$

$\therefore$ motion is simple harmonic with period of $\frac{\pi}{2}$ and centre at $x = 1$.

- (ii) What distance does the particle travel during a full period of its motion? 2

At the extremities, $v = 0$

$$0 = 128 + 32x - 16x^2$$

$$x^2 - 2x - 8 = 0$$

$$(x-4)(x+2) = 0$$

$\therefore$ extremities are at $x = -2, 4$

$$\therefore A = 3$$

The particle will travel a distance of $4A = 12\text{m}$ during a full period of its motion.

- (d) The position vectors of points A and B are $\begin{pmatrix} -7 \\ -3 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 5 \\ -1 \\ -8 \end{pmatrix}$ respectively. 3

Find the vector equation of the sphere with AB as its diameter.

Position vector of centre:
$$\begin{pmatrix} \frac{-7+5}{2} \\ \frac{-3+(-1)}{2} \\ \frac{2+(-8)}{2} \end{pmatrix} = \begin{pmatrix} -1 \\ -2 \\ -3 \end{pmatrix}$$

$$\overrightarrow{AB} = \begin{pmatrix} 5-(-7) \\ -1-(-3) \\ -8-2 \end{pmatrix} = \begin{pmatrix} 12 \\ 2 \\ -10 \end{pmatrix}$$

Radius of sphere:
$$\frac{1}{2}|\overrightarrow{AB}| = \frac{1}{2}\sqrt{12^2 + 2^2 + (-10)^2} = \sqrt{62}$$

Vector equation of sphere:

$$\left| \underline{r} - \begin{pmatrix} -1 \\ -2 \\ -3 \end{pmatrix} \right| = \sqrt{62}$$

- (e) (i) Prove that $\forall x, y \in \mathbb{R}^+, \frac{x+y}{2} \geq \sqrt{xy}$. 1

$$\begin{aligned} (a-b)^2 &\geq 0 \\ a^2 - 2ab + b^2 &\geq 0 \\ a^2 + b^2 &\geq 2ab \\ \frac{a^2 + b^2}{2} &\geq ab \end{aligned}$$

Sub $a = \sqrt{x}, b = \sqrt{y}$

$$\frac{x+y}{2} \geq \sqrt{xy}$$

- (ii) Let a, b, c and d be positive real numbers. Use the result from (i) to prove that 2

$$(a+b+c+d)\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d}\right) \geq 16.$$

Method 1:
Using (i)

$$\begin{aligned} a+b &\geq 2\sqrt{ab}, \quad c+d \geq 2\sqrt{cd}, \quad \frac{1}{a} + \frac{1}{b} \geq 2\sqrt{\frac{1}{ab}}, \quad \frac{1}{c} + \frac{1}{d} \geq 2\sqrt{\frac{1}{cd}} \\ \therefore 2\sqrt{ab}, 2\sqrt{cd}, \frac{2}{\sqrt{ab}}, \frac{2}{\sqrt{cd}} &> 0 \end{aligned}$$

$$\begin{aligned}
LHS &= (a+b+c+d) \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d} \right) \\
&\geq (2\sqrt{ab} + 2\sqrt{cd}) \left(\frac{2}{\sqrt{ab}} + \frac{2}{\sqrt{cd}} \right) \\
&= 4 + \frac{4\sqrt{ab}}{\sqrt{cd}} + \frac{4\sqrt{cd}}{\sqrt{ab}} + 4 \\
&= 8 + 4 \left(\frac{\sqrt{ab}}{\sqrt{cd}} + \frac{\sqrt{cd}}{\sqrt{ab}} \right) \\
&\geq 8 + 4 \left(2\sqrt{\frac{\sqrt{abcd}}{\sqrt{abcd}}} \right) \\
&= 16
\end{aligned}$$

Method 2:

$$\begin{aligned}
LHS &= (a+b+c+d) \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d} \right) \\
&= 1 + \frac{a}{b} + \frac{a}{c} + \frac{a}{d} + \frac{b}{a} + 1 + \frac{b}{c} + \frac{b}{d} + \frac{c}{a} + \frac{c}{b} + 1 + \frac{c}{d} + \frac{d}{a} + \frac{d}{b} + \frac{d}{c} + 1 \\
&= 4 + \left(\frac{a}{b} + \frac{b}{a} \right) + \left(\frac{a}{c} + \frac{c}{a} \right) + \left(\frac{a}{d} + \frac{d}{a} \right) + \left(\frac{b}{c} + \frac{c}{b} \right) + \left(\frac{b}{d} + \frac{d}{b} \right) + \left(\frac{c}{d} + \frac{d}{c} \right) \\
&\geq 4 + 2\sqrt{\frac{ab}{ab}} + 2\sqrt{\frac{ac}{ac}} + 2\sqrt{\frac{ad}{ad}} + 2\sqrt{\frac{bc}{bc}} + 2\sqrt{\frac{bd}{bd}} + 2\sqrt{\frac{cd}{cd}} \quad (\text{using (i)}) \\
&= 16
\end{aligned}$$

Question 13 (15 marks) Use a SEPARATE writing booklet

(a) Evaluate $\int_0^{\pi} x \sin x \cos^4 x \, dx$ using the property $\int_0^a f(x) \, dx = \int_0^a f(a-x) \, dx$.

2

$$\begin{aligned} & \int_0^{\pi} x \sin x \cos^4 x \, dx \\ &= \int_0^{\pi} (\pi - x) \sin(\pi - x) \cos^4(\pi - x) \, dx \\ &= \int_0^{\pi} (\pi - x) \sin x (-\cos x)^4 \, dx \\ &= \int_0^{\pi} (\pi - x) \sin x \cos^4 x \, dx \\ &= \pi \int_0^{\pi} \sin x \cos^4 x \, dx - \int_0^{\pi} x \sin x \cos^4 x \, dx \end{aligned}$$

$$\begin{aligned} 2 \int_0^{\pi} x \sin x \cos^4 x \, dx &= \pi \int_0^{\pi} \sin x \cos^4 x \, dx \\ \int_0^{\pi} x \sin x \cos^4 x \, dx &= \frac{\pi}{2} \int_0^{\pi} \sin x \cos^4 x \, dx \end{aligned}$$

$$\begin{aligned} & \int_0^{\pi} x \sin x \cos^4 x \, dx \\ &= \frac{\pi}{2} \left[\frac{-\cos^5 x}{5} \right]_0^{\pi} \\ &= \frac{\pi}{2} \left(\frac{-\cos^5 \pi}{5} - \left(\frac{-\cos^5 0}{5} \right) \right) \\ &= \frac{\pi}{5} \end{aligned}$$

(b) Consider the two lines in three dimensions given by

3

$$\vec{r} = \begin{pmatrix} -2 \\ 5 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 3 \\ -6 \end{pmatrix} \text{ and } \vec{r} = \begin{pmatrix} 5 \\ 4 \\ -1 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ -5 \\ 8 \end{pmatrix}.$$

Find the point of intersection of the two lines.

Equating $\hat{i}$ components

$$-2 + \lambda = 5 + 2\mu \quad (1)$$

Equating $\hat{j}$ components

$$5 + 3\lambda = 4 - 5\mu \quad (2)$$

(2) $- 3 \times$ (1):

$$(5 + 3\lambda) - 3(-2 + \lambda) = (4 - 5\mu) - 3(5 + 2\mu)$$

$$11 = -11 - 11\mu$$

$$\mu = -2$$

Sub back into (1)

$$\lambda = 3$$

Check using $\hat{k}$ component:

$$1 - 6(3) = -17$$

$$-1 + 8(-2) = -17$$

$$\text{Intersection point: } (-2 + 3, 5 + 3(3), -17) = (1, 14, -17)$$

- (c) The acceleration of a particle moving in a straight line is given by $a = 4 - v^2$, where v is its velocity.

Initially, the particle has a velocity of $\frac{2}{3} \text{ ms}^{-1}$ at $x = 0$.

- (i) What is the displacement of the particle when it is travelling at a velocity of 1 ms^{-1} ?

2

Method 1 Indefinite integral approach:

$$\begin{aligned} v \frac{dv}{dx} &= 4 - v^2 \\ \int \frac{v dv}{4 - v^2} &= \int dx \\ -\frac{1}{2} \ln(4 - v^2) &= x + C \end{aligned}$$

When $x = 0$, $v = \frac{2}{3}$

$$\begin{aligned} -\frac{1}{2} \ln\left(4 - \left(\frac{2}{3}\right)^2\right) &= 0 + C \\ C &= -\frac{1}{2} \ln\left(\frac{32}{9}\right) \\ -\frac{1}{2} \ln(4 - v^2) &= x - \frac{1}{2} \ln\left(\frac{32}{9}\right) \end{aligned}$$

When $v = 1$

$$\begin{aligned} -\frac{1}{2} \ln(4 - 1^2) &= x - \frac{1}{2} \ln\left(\frac{32}{9}\right) \\ x &= \frac{1}{2} \ln\left(\frac{32}{9}\right) - \frac{1}{2} \ln 3 \\ x &= \frac{1}{2} \ln \frac{32}{27} \end{aligned}$$

Method 2 Definite integral approach:

$$\begin{aligned} v \frac{dv}{dx} &= 4 - v^2 \\ \int_{\frac{2}{3}}^1 \frac{v dv}{4 - v^2} &= \int_0^x dx \\ \left[-\frac{1}{2} \ln(4 - v^2) \right]_{\frac{2}{3}}^1 &= [x]_0^x \\ -\frac{1}{2} \ln(4 - 1^2) + \frac{1}{2} \ln\left(4 - \left(\frac{2}{3}\right)^2\right) &= x \\ x &= \frac{1}{2} \left(\ln \frac{32}{9} - \ln 3 \right) \\ x &= \frac{1}{2} \ln \frac{32}{27} \end{aligned}$$

- (ii) What is the velocity of the particle at $t = \frac{1}{2}$?

$$\frac{dv}{dt} = 4 - v^2$$

$$\int \frac{dv}{4 - v^2} = \int dt$$

$$\int \frac{dv}{(2 - v)(2 + v)} = \int dt$$

By splitting into partial fractions:

$$\int \frac{1}{4(2 - v)} + \frac{1}{4(2 + v)} dv = \int dt$$

$$\frac{1}{4} [-\ln(2 - v) + \ln(2 + v)] = t + C_1$$

$$\frac{1}{4} [-\ln(2 - v) + \ln(2 + v)] = t + C_1$$

$$\ln\left(\frac{2 + v}{2 - v}\right) = 4t + C_2$$

When $t = 0$, $v = \frac{2}{3}$

$$\ln\left(\frac{2 + \frac{2}{3}}{2 - \frac{2}{3}}\right) = 4(0) + C_2$$

$$C_2 = \ln 2$$

$$\ln\left(\frac{2 + v}{2 - v}\right) = 4t + \ln 2$$

$$\frac{2 + v}{2 - v} = 2e^{4t}$$

$$2 + v = (2 - v)2e^{4t}$$

$$(1 + 2e^{4t})v = 2(2e^{4t} - 1)$$

$$v = \frac{2(2e^{4t} - 1)}{2e^{4t} + 1}$$

When $t = \frac{1}{2}$

$$v = \frac{2\left(2e^{4\left(\frac{1}{2}\right)} - 1\right)}{2e^{4\left(\frac{1}{2}\right)} + 1} = \frac{2(2e^2 - 1)}{(2e^2 + 1)} \text{ ms}^{-1}$$

(d) The integral I_n is defined by $I_n = \int_0^1 e^{-x} (1-x)^n dx$ for $n \geq 0$.

(i) Show that $I_n = 1 - nI_{n-1}$ for $n \geq 1$.

2

By using integration by parts:

$$\begin{aligned}
 I_n &= \int_0^1 e^{-x} (1-x)^n dx \\
 &= - \int_0^1 (1-x)^n d(e^{-x}) \\
 &= - \left[(1-x)^n e^{-x} \right]_0^1 + \int_0^1 e^{-x} d((1-x)^n) \\
 &= - \left[(1-1)^n e^{-1} - (1-0)^n e^{-0} \right] + \int_0^1 e^{-x} (-n(1-x)^{n-1}) dx \\
 &= 1 - n \int_0^1 e^{-x} (1-x)^{n-1} dx \\
 &= 1 - nI_{n-1}
 \end{aligned}$$

(ii) Given that I_n can be expressed in the form $a + be^{-1}$ where a and b are integers, find an expression for b in terms of n .

2

$$\begin{aligned}
 I_0 &= 1 - e^{-1} \\
 I_1 &= e^{-1} \\
 I_2 &= 1 - 2e^{-1} \\
 I_3 &= 1 - 3(1 - 2e^{-1}) = -2 + 3 \times 2e^{-1} \\
 I_4 &= 1 - 4(-2 + 3 \times 2e^{-1}) = 9 - 4 \times 3 \times 2e^{-1} \\
 &\vdots \\
 I_n &= a + [(-1)^{n+1} n!] e^{-1} \\
 \therefore b &= (-1)^{n+1} n!
 \end{aligned}$$

Question 14 (16 marks) Use a SEPARATE writing booklet

- (a) Prove by mathematical induction that $n^3 - 1 > n^2 + n$, for integers $n \geq 2$.

3

Prove that it is true for $n = 2$

When $n = 2$,

$$LHS = 2^3 - 1 = 7$$

$$RHS = 2^2 + 2 = 6$$

$$\therefore LHS > RHS$$

$\therefore$ it is true for $n = 2$

Assume that it is true for $n = k$

$$k^3 - 1 > k^2 + k$$

Prove that it is true for $n = k + 1$

$$\text{i.e. RTP: } (k+1)^3 - 1 > (k+1)^2 + (k+1)$$

$$(k+1)^3 - 1 - (k+1)^2 - (k+1) > 0$$

$$\begin{aligned} LHS &= (k+1)^3 - 1 - (k+1)^2 - (k+1) \\ &= k^3 + 3k^2 + 3k + 1 - 1 - (k^2 + 2k + 1) - k - 1 \\ &= (k^3 - 1) + 2k^2 - 1 \\ &> (k^2 + k) + 2k^2 - 1 \quad (\text{by assumption}) \\ &= 3k^2 + k - 1 \\ &> k - 1 \quad (\because 3k^2 > 0) \\ &> 1 \quad (\because k \geq 2) \\ &> 0 \end{aligned}$$

$\therefore$ it is true for $n = k + 1$ if it is true for $n = k$

Hence the formula is true by mathematical induction.

(b) Use the method of partial fractions to find the value of a , where $a > 0$, such that

4

$$\int_0^a \frac{3-5x}{(1+x)(1-x+2x^2)} dx = 0.$$

$$\frac{3-5x}{(1+x)(1-x+2x^2)} = \frac{A}{1+x} + \frac{Bx+C}{1-x+2x^2}$$

$$3-5x = A(1-x+2x^2) + (Bx+C)(1+x)$$

Sub $x = -1$

$$3-5(-1) = A(1-(-1)+2(-1)^2)$$

$$A = 2$$

Sub $x = 0$

$$3-5(0) = A+C$$

$$C = 1$$

Sub $x = 1$

$$3-5(1) = A(1-1+2(1)^2) + (B+C)(1+1)$$

$$-2 = 2A + 2(B+C)$$

$$B = -4$$

$$\int_0^a \frac{3-5x}{(1+x)(1-x+2x^2)} dx = \int_0^a \frac{2}{1+x} + \frac{-4x+1}{1-x+2x^2} dx = 0$$

$$\int_0^a \frac{2}{1+x} + \frac{-4x+1}{1-x+2x^2} dx = 0$$

$$\left[2 \ln|1+x| - \ln|1-x+2x^2| \right]_0^a = 0$$

$$(2 \ln|1+a| - \ln|1-a+2a^2|) - (2 \ln|1+0| - \ln|1-0+2 \times 0^2|) = 0$$

$$2 \ln|1+a| - \ln|1-a+2a^2| = 0$$

$$\ln \left| \frac{(1+a)^2}{1-a+2a^2} \right| = 0$$

$$\frac{(1+a)^2}{1-a+2a^2} = \pm 1$$

$$(1+a)^2 = 1-a+2a^2$$

$$1+2a+a^2 = 1-a+2a^2$$

$$a^2 - 3a = 0$$

$$a(a-3) = 0$$

$$a = 0, 3$$

$$(1+a)^2 = -(1-a+2a^2)$$

$$1+2a+a^2 = -1+a-2a^2$$

$$3a^2 + a + 2 = 0$$

no solution as $\Delta < 0$

$$\therefore a > 0, a = 3$$

(c) Let $w = \cos \theta + i \sin \theta$, where $0 < \theta < \pi$.

It is known that the complex number $w^2 + \frac{3\sqrt{2}}{w} - 3$ is purely imaginary.

(i) Show that $2 \cos^2 \theta + 3\sqrt{2} \cos \theta - 4 = 0$.

2

$$\begin{aligned}
 & w^2 + \frac{3\sqrt{2}}{w} - 3 \\
 &= (\cos \theta + i \sin \theta)^2 + 3\sqrt{2} (\cos \theta + i \sin \theta)^{-1} - 3 \\
 &= (\cos 2\theta + i \sin 2\theta) + 3\sqrt{2} (\cos(-\theta) + i \sin(-\theta)) - 3 \\
 &= (\cos 2\theta + 3\sqrt{2} \cos \theta - 3) + i (\sin 2\theta - 3\sqrt{2} \sin \theta) \\
 & \quad \operatorname{Re} \left(w^2 + \frac{3\sqrt{2}}{w} - 3 \right) = 0 \\
 & \quad \cos 2\theta + 3\sqrt{2} \cos \theta - 3 = 0 \\
 & \quad 2 \cos^2 \theta - 1 + 3\sqrt{2} \cos \theta - 3 = 0 \\
 & \quad 2 \cos^2 \theta + 3\sqrt{2} \cos \theta - 4 = 0
 \end{aligned}$$

(ii) Hence, or otherwise, find w .

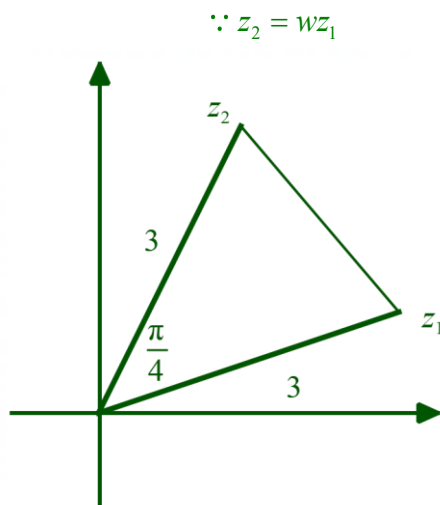
2

$$\begin{aligned}
 & 2 \cos^2 \theta + 3\sqrt{2} \cos \theta - 4 = 0 \\
 & \cos \theta = \frac{-3\sqrt{2} \pm \sqrt{(3\sqrt{2})^2 - 4(2)(-4)}}{2(2)} \\
 & \cos \theta = \frac{-3\sqrt{2} \pm 5\sqrt{2}}{4} \\
 & \cos \theta = \frac{\sqrt{2}}{2}, \quad \cos \theta = -2\sqrt{2} \\
 & \cos \theta = -2\sqrt{2} \text{ has no solution as } -2\sqrt{2} < -1
 \end{aligned}$$

$$\begin{aligned}
 & \cos \theta = \frac{\sqrt{2}}{2} \\
 & \theta = \frac{\pi}{4} \text{ as } 0 < \theta < \pi \\
 & w = \operatorname{cis} \frac{\pi}{4}
 \end{aligned}$$

- (iii) A and B are two points on an Argand diagram representing two distinct complex numbers z_1 and z_2 respectively. Suppose that $z_2 = wz_1$ and $|z_1| = 3$. Find the exact length of AB .

Method 1: Geometrical approach



$$AB^2 = 3^2 + 3^2 - 2 \times 3 \times 3 \times \cos \frac{\pi}{4}$$

$$AB^2 = 18 - 9\sqrt{2}$$

$$AB = 3\sqrt{2 - \sqrt{2}}$$

Method 2: Algebraic approach

$$\begin{aligned} AB &= |z_2 - z_1| \\ &= |wz_1 - z_1| \\ &= |z_1(w - 1)| \\ &= |z_1||w - 1| \\ &= 3 \left| \cos \frac{\pi}{4} + i \sin \frac{\pi}{4} - 1 \right| \\ &= 3 \left| \left(\cos \frac{\pi}{4} - 1 \right) + i \sin \frac{\pi}{4} \right| \\ &= 3 \sqrt{\left(\cos \frac{\pi}{4} - 1 \right)^2 + \sin^2 \frac{\pi}{4}} \\ &= 3 \sqrt{-2 \cos \frac{\pi}{4} + 1 + 1} \\ &= 3 \sqrt{2 - 2 \left(\frac{\sqrt{2}}{2} \right)} \\ &= 3\sqrt{2 - \sqrt{2}} \end{aligned}$$

(d) Consider $f(x) = x^m - mx$ where $m > 1$.

By using calculus, or otherwise, show that $x^m - 1 \geq m(x - 1)$ for all $x > 0$.

$$f'(x) = mx^{m-1} - m$$

When $f'(x) = 0$

$$0 = mx^{m-1} - m$$

$$m(x^{m-1} - 1) = 0$$

$$x^{m-1} = 1$$

$$x = 1 \quad (\because x > 0)$$

$$f''(x) = m(m-1)x^{m-2}$$

$$f''(1) = m(m-1) > 0 \quad (\because m > 1)$$

$\therefore$ at $x = 1$, there is a min turning point

$$f(1) = 1^m - m(1) = 1 - m$$

$$f(x) > 1 - m \quad (\because \text{there are no other turning points})$$

$$x^m - mx > 1 - m$$

$$x^m - 1 > mx - m$$

$$x^m - 1 > m(x - 1)$$

Question 15 (14 marks) Use a SEPARATE writing booklet

- (a) Let z and w be non-zero complex numbers. Consider the statements below:

3

$$P: z\bar{w} + \bar{z}w = 0$$

$$Q: \frac{z}{w} = ik, k \in \mathbb{R}$$

Prove that $P \Rightarrow Q$.

Method 1:

Prove $P \Rightarrow Q$ i.e. if $z\bar{w} + \bar{z}w = 0$, then $\frac{z}{w} = ik$

$$z\bar{w} + \bar{z}w = 0$$

$$z\bar{w} + \overline{z\bar{w}} = 0$$

$$2\operatorname{Re}(z\bar{w}) = 0$$

$\therefore z\bar{w}$ is purely imaginary

$$z\bar{w} = ia, a \in \mathbb{R}$$

$$\frac{z\bar{w}w}{w} = ia$$

$$\frac{z|w|^2}{w} = ia$$

$$\frac{z}{w} = i \frac{a}{|w|^2}$$

$$\frac{z}{w} = ik, k \in \mathbb{R} \left(\because a, |w|^2 \in \mathbb{R} \right)$$

Method 2:

Let $z = a + bi, w = c + di$

$$z\bar{w} + \bar{z}w = 0$$

$$(a + bi)(c - di) + (a - bi)(c + di) = 0$$

$$ac + bd + i(bc - ad) + ac + bd + i(-bc + ad) = 0$$

$$2(ac + bd) = 0$$

$$ac + bd = 0$$

z and w can also be considered as two vectors $\begin{pmatrix} a \\ b \end{pmatrix}, \begin{pmatrix} c \\ d \end{pmatrix}$, hence the above results is equivalent to

$\begin{pmatrix} a \\ b \end{pmatrix} \cdot \begin{pmatrix} c \\ d \end{pmatrix} = 0$, i.e. $\begin{pmatrix} a \\ b \end{pmatrix} \perp \begin{pmatrix} c \\ d \end{pmatrix}$, this means z and w must be perpendicular on the argand diagram,

$$z = (ik)w, k \in \mathbb{R}$$

$$\frac{z}{w} = ik$$

(b) (i) By using a suitable substitution, show that

3

$$\int \sin \sqrt{x} \, dx = 2 \sin \sqrt{x} - 2\sqrt{x} \cos \sqrt{x} + C.$$

$$\text{Let } u = \sqrt{x}$$

$$\frac{du}{dx} = \frac{1}{2\sqrt{x}}$$

$$\frac{du}{dx} = \frac{1}{2u}$$

$$2u \, du = dx$$

$$\int \sin \sqrt{x} \, dx = \int \sin u \, (2u \, du) = 2 \int u \sin u \, du$$

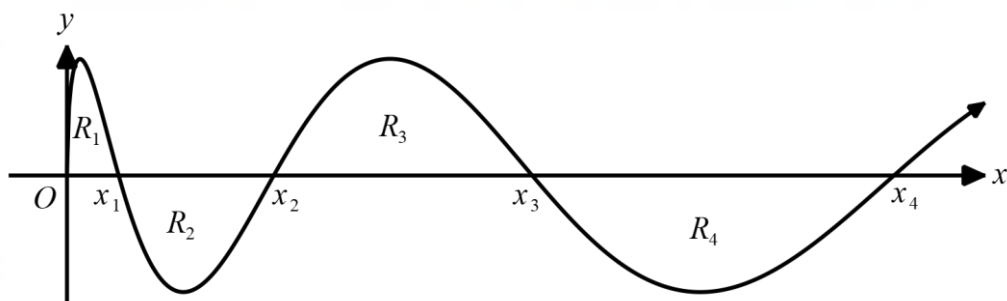
Using integration by parts:

$$2 \int u \sin u \, du = -2 \int u \, d(\cos u) = -2 \left(u \cos u - \int \cos u \, du \right) = -2u \cos u + 2 \sin u + C$$

Sub $u = \sqrt{x}$:

$$\int \sin \sqrt{x} \, dx = -2\sqrt{x} \cos \sqrt{x} + 2 \sin \sqrt{x} + C$$

The diagram below shows part of the curve $y = \sin \sqrt{x}$ for $x \geq 0$.



The curve intersects the x -axis at $0, x_1, x_2, x_3, x_4, \dots$

The regions bounded by the curve and the x -axis are defined by $R_1, R_2, R_3, R_4, \dots$ as shown in the diagram.

(ii) By first finding an expression for x_n , or otherwise, express the area of the region R_n in terms of n and π .

3

$$\sin \sqrt{x} = 0$$

$$\sqrt{x} = 0, \pi, 2\pi, \dots$$

$$\sqrt{x} = n\pi$$

$$x = n^2 \pi^2$$

$$\begin{aligned}
R_n &= \left| \int_{x_{n-1}}^{x_n} \sin \sqrt{x} \, dx \right| \\
&= (-1)^{n+1} \int_{(n-1)^2 \pi^2}^{n^2 \pi^2} \sin \sqrt{x} \, dx \\
(-1)^{n+1} R_n &= \left[-2\sqrt{x} \cos \sqrt{x} + 2 \sin \sqrt{x} \right]_{(n-1)^2 \pi^2}^{n^2 \pi^2} \\
&= \left(-2\sqrt{n^2 \pi^2} \cos \sqrt{n^2 \pi^2} + 2 \sin \sqrt{n^2 \pi^2} \right) - \left(-2\sqrt{(n-1)^2 \pi^2} \cos \sqrt{(n-1)^2 \pi^2} + 2 \sin \sqrt{(n-1)^2 \pi^2} \right) \\
&= -2n\pi \cos n\pi + 2 \sin n\pi + 2(n-1)\pi \cos(n-1)\pi - 2 \sin(n-1)\pi \\
&= -2n\pi \cos n\pi + 2(n-1)\pi \cos(n-1)\pi \quad (\because \sin n\pi = 0) \\
&= -2n\pi \cos n\pi + 2(n-1)\pi (\cos n\pi \cos \pi + \sin n\pi \sin \pi) \\
&= -2n\pi \cos n\pi - 2(n-1)\pi \cos n\pi \\
&= (-4n+2)\pi \cos n\pi \\
&= (-4n+2)\pi (-1)^n \quad (\because \cos n\pi = -1 \text{ if } n \text{ is odd and } \cos n\pi = 1 \text{ if } n \text{ is even}) \\
R_n &= -(-4n+2)\pi \\
&= (4n-2)\pi
\end{aligned}$$

Question 15 continues on page 13

Question 15 (continued)

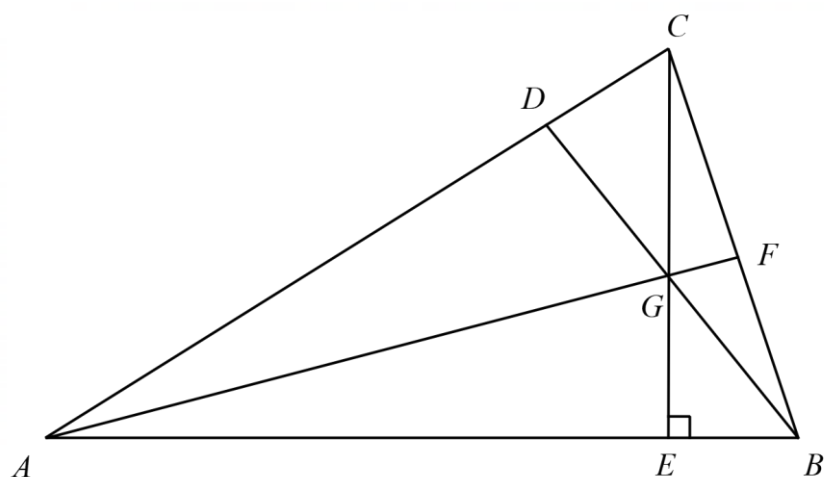
(c) The diagram shows triangle ABC .

The point E lies on AB so that $AE : EB = \lambda : 1$ and $CE \perp AB$.

The point G is the mid-point of CE .

BG is produced to meet AC at D and $AD : DC = r : 1$.

The point F is chosen on BC so that A, G, F are collinear.



NOT TO
SCALE

Let $\overrightarrow{BE} = \underline{u}$ and $\overrightarrow{EG} = \underline{v}$.

(i) Express $\overrightarrow{BG}$ in terms of $\underline{u}$ and $\underline{v}$.

$$\overrightarrow{BG} = \underline{u} + \underline{v}$$

1

(ii) By considering $\overrightarrow{BD}$, use vector methods to show that $r = \lambda + 1$.

4

$$\overrightarrow{BD} = k \cdot \overrightarrow{BG} = k(\underline{u} + \underline{v}), \quad k \in \mathbb{R} \quad (\text{as } B, G, D \text{ collinear}) \quad (1)$$

$$\begin{aligned} \overrightarrow{BD} &= \overrightarrow{BA} + \overrightarrow{AD} \\ &= (\lambda + 1)\overrightarrow{BE} + \frac{r}{r+1}\overrightarrow{AC} \\ &= (\lambda + 1)\underline{u} + \frac{r}{r+1}(\overrightarrow{AE} + \overrightarrow{EC}) \\ &= (\lambda + 1)\underline{u} + \frac{r}{r+1}(-\lambda\overrightarrow{BE} + 2\overrightarrow{EG}) \\ &= (\lambda + 1)\underline{u} + \frac{r}{r+1}(-\lambda\underline{u} + 2\underline{v}) \\ &= \left(\lambda + 1 - \frac{r\lambda}{r+1}\right)\underline{u} + \frac{2r}{r+1}\underline{v} \quad (2) \end{aligned}$$

Equating coefficients of $\underline{u}$ and $\underline{v}$ in (1) and (2) [as $\underline{u}$ and $\underline{v}$ are independent]:

$$\begin{aligned} \lambda + 1 - \frac{r\lambda}{r+1} &= \frac{2r}{r+1} \quad (=k) \\ (\lambda + 1)(r+1) - r\lambda &= 2r \\ \cancel{\lambda r} + \lambda + r + 1 - \cancel{r\lambda} &= 2r \\ r &= \lambda + 1 \end{aligned}$$

End of Question 15

Question 16 (16 marks) Use a SEPARATE writing booklet

- (a) It is given that for positive numbers $a_1, a_2, a_3, \dots, a_n$ with arithmetic mean A ,

$$A^n \geq a_1 \times a_2 \times a_3 \times \dots \times a_n. \quad (\text{Do NOT prove this.})$$

Let k be a positive integer.

- (i) Prove that $k^k \geq 1 \times 3 \times 5 \times \dots \times (2k-1)$.

2

Consider the positive odd numbers $1, 3, 5, \dots, (2k-1)$

Using the results given:

$$\left[\frac{(1+3+5+\dots+(2k-1))}{k} \right]^k \geq 1 \times 3 \times 5 \times \dots \times (2k-1)$$

Using sum of AP on the LHS

$$\left[\frac{\frac{k}{2}(1+(2k-1))}{k} \right]^k \geq 1 \times 3 \times 5 \times \dots \times (2k-1)$$

$$\left[\frac{1}{2}(2k) \right]^k \geq 1 \times 3 \times 5 \times \dots \times (2k-1)$$

$$k^k \geq 1 \times 3 \times 5 \times \dots \times (2k-1)$$

- (ii) Hence prove that $(k^2 + k)^k \geq (2k)!$.

2

Consider the positive even numbers $2, 4, 6, \dots, 2k$

Using the results given:

$$\left[\frac{(2+4+6+\dots+2k)}{k} \right]^k \geq 2 \times 4 \times 6 \times \dots \times 2k$$

Using sum of AP on the LHS

$$\left[\frac{\frac{k}{2}(2+2k)}{k} \right]^k \geq 2 \times 4 \times 6 \times \dots \times 2k$$

$$\left[\frac{1}{2}(2k+2) \right]^k \geq 2 \times 4 \times 6 \times \dots \times 2k$$

$$(k+1)^k \geq 2 \times 4 \times 6 \times \dots \times 2k$$

As LHS and RHS of both inequalities are positive, multiplying the two inequalities give:

$$k^k (k+1)^k \geq 1 \times 3 \times 5 \times \dots \times (2k-1) \times 2 \times 4 \times 6 \times \dots \times 2k$$

$$(k(k+1))^k \geq 1 \times 2 \times 3 \times \dots \times (2k-1) \times 2k$$

$$(k^2 + k)^k \geq (2k)!$$

(b) Consider the equation $(z+2)^8 = z^8$.

(i) Show that the 6 non-real roots of the equation can be expressed in the form

4

$$-1 - i \cot\left(\frac{k\pi}{8}\right), \quad k = \pm 1, \pm 2, \pm 3.$$

$$\left(\frac{z+2}{z}\right)^8 = 1$$

$$\left(\frac{z+2}{z}\right)^8 = \text{cis } 0$$

$$\frac{z+2}{z} = \text{cis}\left(\frac{2k\pi}{8}\right), \quad k = \pm 1, \pm 2, \pm 3, \pm 4$$

When $k = \pm 4$, $\text{cis}\left(\frac{2(\pm 4)\pi}{8}\right) = \text{cis}(\pm\pi) = -1$ (which is a real root)

The 6 non-real roots must be

$$\frac{z+2}{z} = \text{cis}\left(\frac{2k\pi}{8}\right), \quad k = \pm 1, \pm 2, \pm 3$$

$$z+2 = z \text{cis}\left(\frac{2k\pi}{8}\right)$$

$$\left(\text{cis}\left(\frac{2k\pi}{8}\right) - 1\right)z = 2$$

$$z = \frac{2}{\text{cis}\left(\frac{2k\pi}{8}\right) - 1}$$

$$\begin{aligned}
z &= \frac{2}{\operatorname{cis}\left(\frac{2k\pi}{8}\right) - 1} \\
z &= \frac{2}{\left(\cos\left(\frac{2k\pi}{8}\right) - 1\right) + i\sin\left(\frac{2k\pi}{8}\right)} \\
z &= \frac{2\left(\left(\cos\left(\frac{2k\pi}{8}\right) - 1\right) - i\sin\left(\frac{2k\pi}{8}\right)\right)}{\left(\cos\left(\frac{2k\pi}{8}\right) - 1\right)^2 + \sin^2\left(\frac{2k\pi}{8}\right)} \\
z &= \frac{2\left(-\sin^2\left(\frac{k\pi}{8}\right)\right) - i\left(2\sin\left(\frac{k\pi}{8}\right)\cos\left(\frac{k\pi}{8}\right)\right)}{\cos^2\left(\frac{2k\pi}{8}\right) - 2\cos\left(\frac{2k\pi}{8}\right) + 1 + \sin^2\left(\frac{2k\pi}{8}\right)} \\
z &= \frac{-2\sin\left(\frac{k\pi}{8}\right)\left(\sin\left(\frac{k\pi}{8}\right) + i\cos\left(\frac{k\pi}{8}\right)\right)}{2 - 2\cos\left(\frac{2k\pi}{8}\right)} \\
z &= \frac{-2\sin\left(\frac{k\pi}{8}\right)\left(\sin\left(\frac{k\pi}{8}\right) + i\cos\left(\frac{k\pi}{8}\right)\right)}{2\sin^2\left(\frac{k\pi}{8}\right)} \\
z &= \frac{-\sin\left(\frac{k\pi}{8}\right) - i\cos\left(\frac{k\pi}{8}\right)}{\sin\left(\frac{k\pi}{8}\right)} \\
z &= -1 - i\cot\left(\frac{k\pi}{8}\right), \quad k = \pm 1, \pm 2, \pm 3
\end{aligned}$$

Alternate solution involves using t -substitution

- (ii) Hence, find the value of $\sin^2\left(\frac{\pi}{8}\right)\sin^2\left(\frac{2\pi}{8}\right)\sin^2\left(\frac{3\pi}{8}\right)$ given that the product of roots of the equation is -16 .

$$\begin{aligned} & \left(-1-i\cot\left(\frac{\pi}{8}\right)\right)\left(-1-i\cot\left(-\frac{\pi}{8}\right)\right) \\ &= \left(-1-i\cot\left(\frac{\pi}{8}\right)\right)\left(-1+i\cot\left(\frac{\pi}{8}\right)\right) \\ &= \left(1+\cot^2\left(\frac{\pi}{8}\right)\right) \\ &= \csc^2\left(\frac{\pi}{8}\right) \end{aligned}$$

Similarly:

$$\begin{aligned} & \left(-1-i\cot\left(\frac{2\pi}{8}\right)\right)\left(-1-i\cot\left(-\frac{2\pi}{8}\right)\right) = \csc^2\left(\frac{2\pi}{8}\right) \\ & \left(-1-i\cot\left(\frac{3\pi}{8}\right)\right)\left(-1-i\cot\left(-\frac{3\pi}{8}\right)\right) = \csc^2\left(\frac{3\pi}{8}\right) \end{aligned}$$

7 roots in total: real root of -1 plus the 6 non-real roots in (i)

Product of roots:

$$\begin{aligned} & \csc^2\left(\frac{\pi}{8}\right)\csc^2\left(\frac{2\pi}{8}\right)\csc^2\left(\frac{3\pi}{8}\right)(-1) = -16 \\ & \frac{1}{\sin^2\left(\frac{\pi}{8}\right)\sin^2\left(\frac{2\pi}{8}\right)\sin^2\left(\frac{3\pi}{8}\right)} = 16 \\ & \sin^2\left(\frac{\pi}{8}\right)\sin^2\left(\frac{2\pi}{8}\right)\sin^2\left(\frac{3\pi}{8}\right) = \frac{1}{16} \end{aligned}$$

Question 16 continues on page 15

Question 16 (continued)

(c) Let $z = \cos \theta + i \sin \theta$.

(i) Find the four values of z such that $\operatorname{Im}(z^2 + \bar{z}) = 0$.

2

$$\begin{aligned} z^2 + \bar{z} &= (\cos \theta + i \sin \theta)^2 + (\cos \theta - i \sin \theta) \\ &= (\cos 2\theta + i \sin 2\theta) + (\cos \theta - i \sin \theta) \\ &= (\cos 2\theta + \cos \theta) + i(\sin 2\theta - \sin \theta) \end{aligned}$$

$$\operatorname{Im}(z^2 + \bar{z}) = 0$$

$$\sin 2\theta - \sin \theta = 0$$

$$2 \sin \theta \cos \theta - \sin \theta = 0$$

$$\sin \theta (2 \cos \theta - 1) = 0$$

$$\sin \theta = 0 \quad \cos \theta = \frac{1}{2}$$

$$\theta = 0, \pi \quad \theta = \frac{-\pi}{3}, \frac{\pi}{3}$$

$$z = \operatorname{cis} 0, \operatorname{cis} \pi, \operatorname{cis} \frac{-\pi}{3}, \operatorname{cis} \frac{\pi}{3}$$

$$z = 1, -1, \operatorname{cis} \frac{-\pi}{3}, \operatorname{cis} \frac{\pi}{3}$$

Let z_1 and z_2 be two of the values of z obtained in (i) such that $\text{Im}(z_1) < 0 < \text{Im}(z_2)$.

Also let $S_n = \sum_{r=1}^n w^r$, where $w = \frac{z_2}{z_1}$, for any positive integer n .

- (ii) Find all positive integers k such that $(S_n)^k + (S_{n+1})^k + (S_{n+2})^k = 2$ for any positive integer n .

3

$$z_1 = \text{cis } \frac{-\pi}{3}, z_2 = \text{cis } \frac{\pi}{3}$$

$$w = \frac{z_2}{z_1} = \frac{\text{cis } \frac{\pi}{3}}{\text{cis } \frac{-\pi}{3}} = \text{cis } \left(\frac{\pi}{3} - \left(\frac{-\pi}{3} \right) \right) = \text{cis } \frac{2\pi}{3}$$

Since $n, n+1, n+2$ are consecutive integers, one must be a multiple of 3, one must leave a remainder of 1 when divided by 3, one must leave a remainder of 2 when divided by 3

Assuming n is a multiple of 3, $n = 3A, A \in \mathbb{R}^+$

$$\text{Also } w = \text{cis } \frac{2\pi}{3} \therefore w^3 = \left(\text{cis } \frac{2\pi}{3} \right)^3 = 1$$

$$w^3 = 1$$

$$w^3 - 1 = 0$$

$$(w-1)(w^2 + w + 1) = 0$$

$$\begin{aligned} S_n &= \sum_{k=1}^n w^k \\ &= w + w^2 + w^3 + \dots + w^n \\ &= \frac{w(w^n - 1)}{w - 1} \\ &= \frac{w(w^{3A} - 1)}{w - 1} \\ &= \frac{w((w^3)^A - 1)}{w - 1} \\ &= \frac{w(1^A - 1)}{w - 1} \quad (\because w^3 = 1) \\ &= 0 \end{aligned}$$

$$\begin{aligned} S_{n+1} &= \sum_{k=1}^{n+1} w^k \\ &= \sum_{k=1}^n w^k + w^{n+1} \\ &= 0 + w^{n+1} \\ &= w^{3A+1} \\ &= w^{3A} \times w \\ &= (w^3)^A \times w \\ &= 1^A \times w \\ &= w \end{aligned}$$

$$\begin{aligned} S_{n+2} &= \sum_{k=1}^{n+2} w^k \\ &= \sum_{k=1}^{n+1} w^k + w^{n+2} \\ &= w + w^{n+2} \\ &= w + w^{3A+2} \\ &= w + w^{3A} \times w^2 \\ &= w + (w^3)^A \times w^2 \\ &= w + w^2 \\ &= -1 \quad (\because 1 + w + w^2 = 0) \end{aligned}$$

$$(S_n)^k + (S_{n+1})^k + (S_{n+2})^k = 2$$

$$0^k + w^k + (-1)^k = 2$$

Now consider cases when k is odd and when k is even

When k is odd,

$$w^k - 1 = 2$$

$$w^k = 3$$

No solution as $|w| = 1$

When k is even,

$$w^k + 1 = 2$$

$$w^k = 1$$

$\because w^3 = 1$, k must be an even multiple of 3 i.e. k is a multiple of 6 or $k = 6B$, $B \in \mathbb{R}^+$

End of paper